

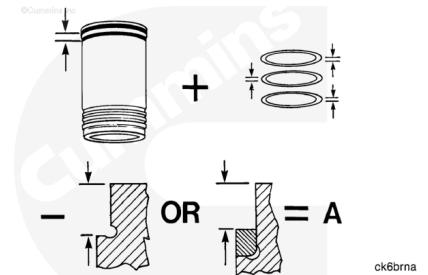
Trainings ▾

Machine

Note : Read the complete procedure before beginning to machine the block.

The counterbore ledge can be machined. Sealing rings will be installed to obtain the correct cylinder liner protrusion.

The cylinder liner protrusion (A) is the total sum of the thickness of the liner flange and the shims, minus the counterbore depth.



Cylinder Liner Protrusion (A)		
mm		in
0.13	MIN	0.005
0.18	MAX	0.007

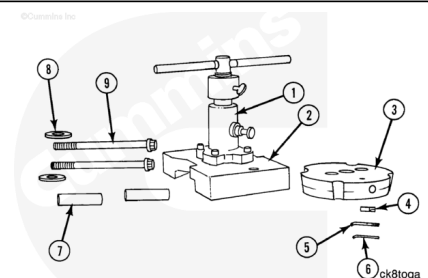
Note : Some liners are oversize in flange thickness and flange outside diameter

If necessary, machine the block for oversize liners before machining the counterbore ledge for depth.

Resurface the block as required before machining the counterbore ledge depth. Reference the Alternative Repair Manual, Bulletin 3379035.

Tools Required:

1. Drive unit, Part Number 3376685
2. Adapter plate, Part Number 3376684
3. Cutter plate, Part Number 3375980
4. Tool bit (part number depends on block style.)
5. Hex wrench [0.1875 inch]
6. Hex wrench [0.0938 inch]
7. Bolt spacers [0.8125 inch inside diameter x 3.500 inch]
8. Plain washers [0.8125 inch inside diameter x 2.000 inch outside diameter]
9. Cylinder head capscrews for the block being serviced.



Note : The cylinder block counterbore tool, Part Number 3376684, contains the tools listed above. The same kit contains the tools to machine the counterbore depth on all Cummins® engines, except the L10.

Install the drive unit (1) on the adapter plate (2).

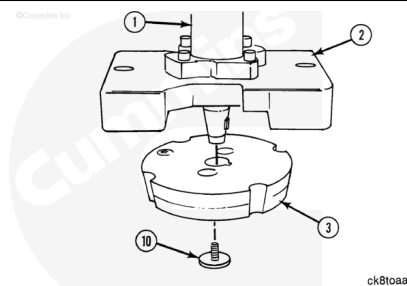
Torque Value: 41 n•m [30 ft-lb]

Align the key slot on the cutter plate (3) with the key on the drive unit shaft. Install the plain washer and the capscrew (10).

Torque Value: 41 n•m [30 ft-lb]

Use a mallet. Tap the cutter plate. The cutter plate **must** be seated on the shaft.

Note : Radius specifications are [inch].



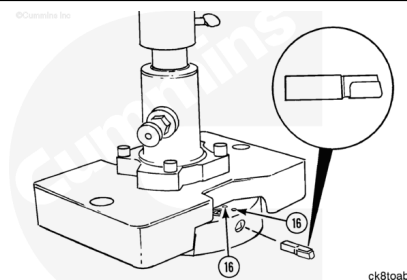
⚠ CAUTION ⚠

Part Number 3377468, must be used to machine the counterbore ledge.

Install the correct tool bit in the cutter plate. Position the bit so it cuts when the plate is moving **clockwise**.

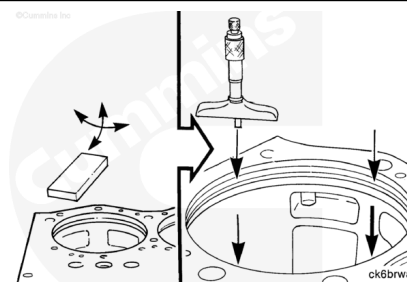
Push the tool bit into the plate until it is even with, or below the outside diameter of the plate.

Some resistance can be felt when installing the bit extension device. If the bit does **not** go in, loosen the two locking screws (16).



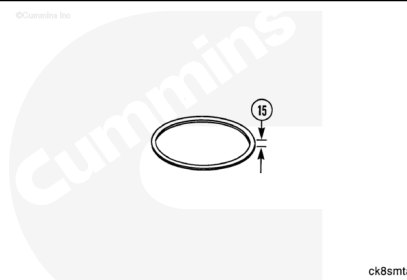
Use a fine India type stone. Remove any nicks and burrs from the head surface and the counterbore ledge. Nicks and burrs prevent the tool from seating correctly.

Use a depth micrometer to measure the counterbore depth. Measure at four equally spaced locations. Record the measurements.



⚠ CAUTION ⚠

Do not use more than three seal rings under one liner. The use of one, thick seal ring is better than two, thin shims.

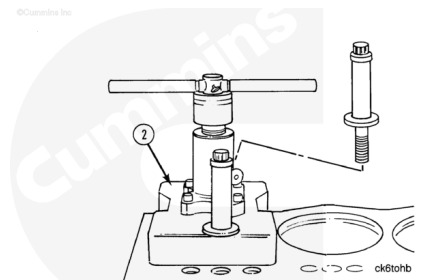


Measure the thickness (15) of the sealing rings used with the engine block being machined.

Select the best combination of shims and counterbore rings to obtain the correct protrusion.

⚠ CAUTION ⚠

Do not allow the weight of the cutter plate to cause it to fall. The bit plate will be damaged. Pull up on the shaft when pulling the locking pin.



Put the cylinder block counterbore tool on the block.

Pull the locking pin. Lower the cutter plate until it touches the counterbore ledge. The taper on the cutter plate centers the tool in the cylinder bore.

There **must** be clearance between the stop collar (17) and the housing. Turn the collar until there is clearance.

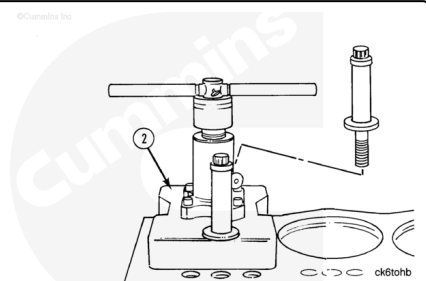
Turn the shaft backward and forward. The taper **must** be positioned squarely in the bore.

There **must** be no clearance between the adapter plate mounting surface and the block head surface.

Align the holes in the adapter plate (2) with two cylinder head capscrew holes in the block. Install the plain washers, spacers, and the cylinder head capscrews.

Torque Value: 70 n•m [52 ft-lb]

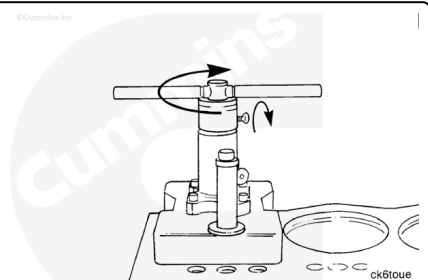
If the tool is **not** aligned correctly, it does **not** turn after torque is applied to the capscrews.



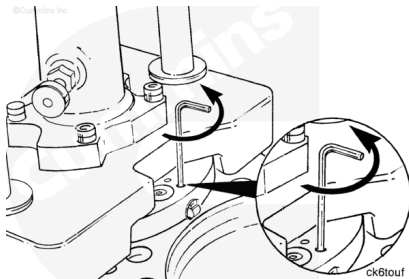
On blocks with a double undercut radius, the tool bit **must** be adjusted so it is in the radius.

Use the stop collar to raise the tool bit from the counterbore ledge 0.025 mm to 0.25 mm [0.001 inch to 0.010 inch].

The tool **must** be raised enough to allow the bit extension device to function.

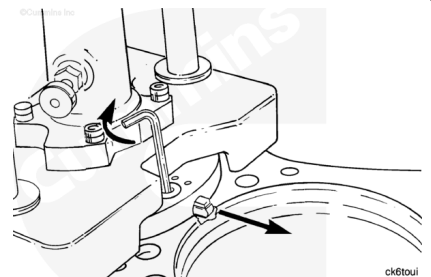


Use a [0.9375 inch] hex wrench. Loosen the tool bit locking screws **only** enough to allow the extension device to function.



Use the [0.1875 inch] hex wrench. Turn the bit extension device until the tool bit touches the inside diameter of the large radius.

This **must** be done to be sure the large radius is extended downward as the ledge is machined.



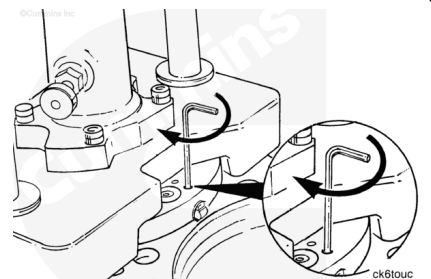
⚠ CAUTION ⚠

Turn the handle counterclockwise. The tool will damage the bore if it is turned the opposite direction.

Use the handle to turn the tool **counterclockwise**. Listen for the sound of the tool bit scraping the bore wall as it is turned a minimum of 180 degrees.

If the sound does **not** continue, use the bit extension device to push the tool bit against the bore wall opposite the first position.

Turn the tool another 180 degrees. Listen again for the sound of the bit. Check for a burr if it touches at one point **only**.

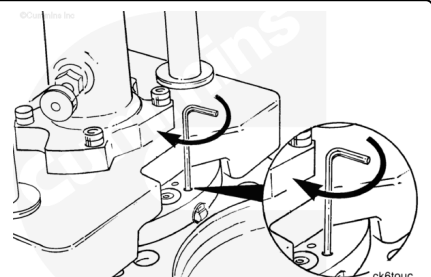


⚠ CAUTION ⚠

Do not attempt to raise the cutter plate with the tool bit extended into the double undercut radius.

Use a [0.9375 inch] hex wrench. Tighten the two locking screws of the tool bit.

Turn the stop collar to lower the cutter plate so that the bit touches the ledge.



Adjust the Depth of Machining

Be sure the tool bit is seated on the counterbore ledge.

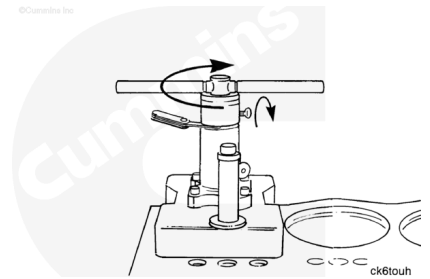
Results will be correct with several 0.05 mm to 0.08 mm [0.002 inch to 0.003 inch] cuts. The tool bit will be damaged by deep cuts. The surface finish will be excessively rough.

Note : Surface finish specification must be 80AA or better.

Be sure the tool bit is **not** raised.

Turn the stop collar **clockwise** to adjust the clearance.

Move a feeler gauge around the housing. Clearance **must** be within specifications at the nearest point.



Tighten the thumb screw on the stop collar.

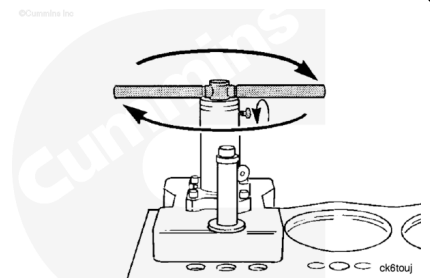
Note : Record the depth of each machining. The figures are required to obtain the correct liner protrusion.

Apply downward pressure. Turn the handle **clockwise**. Keep the tool bit against the ledge.

Machine the bore until the stop collar contacts the housing.

Not using pressure, spin the tool two or three times to smooth the surface.

Machine the counterbore in several operations until the maximum liner protrusion is obtained.



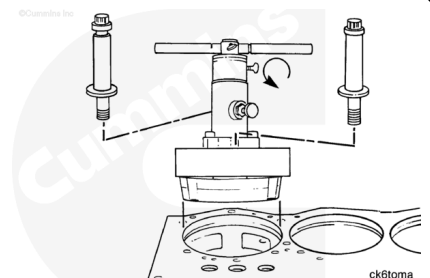
⚠ CAUTION ⚠

Loosen and retract the tool bit before attempting to raise the tool.

Raise the cutter plate.

Remove the tool from the block.

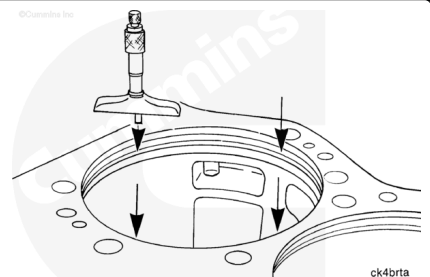
Use a fine India type stone. Remove any burrs and sharp edges from the counterbore ledge.



⚠ CAUTION ⚠

The micrometer must not touch the radius on a block that does not contain a double undercut counterbore radius.

Use a depth micrometer. Measure the counterbore depth in the four locations, as illustrated.



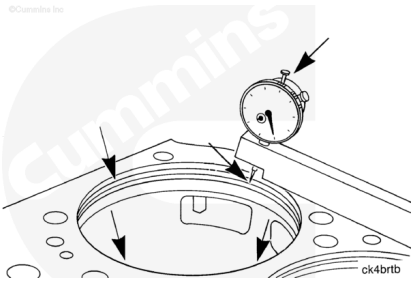
The four measurements **must not** vary more than 0.25 mm [0.01 inch]. If the measurements exceed the specification, the counterbore ledge **must** be machined again.

⚠ CAUTION ⚠

The indicator must not touch the counterbore radius on a block that does not have a double undercut.

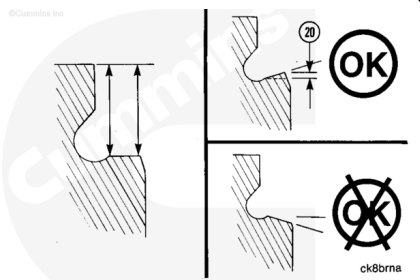
Use gauge block, Part Number 82349, to measure the angle of the counterbore ledge at four, equally spaced locations.

The measurement of the ledge depth **must** be performed as near to the counterbore radius as possible, and as near to the counterbore edge as possible.



The angle (12) is OK if the measurement that is near the counterbore edge is the same or no more than 0.036 mm [0.0014 inch] shorter than the measurement near the counterbore radius.

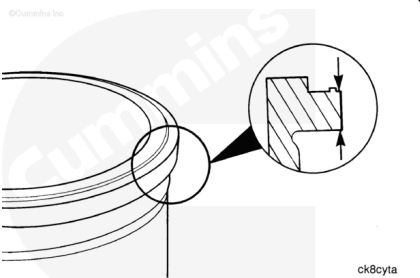
Machine the ledge again if the measurement near the counterbore edge is longer than the measurement near the radius.



Check the counterbore for cracks. Refer to Procedure 001-026 in Section 1 (/qs3/pubsys2/xml/en/procedures/20/20-001-026-tr.html)

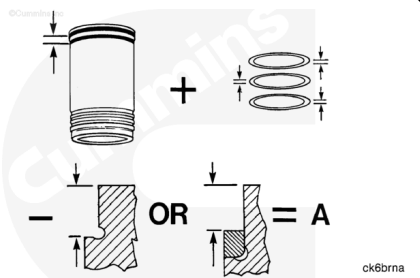
Note : Liners and shims must be identified by the cylinder number for proper assembly.

Measure the liner flange thickness at four, equally spaced locations around the flange.



Cylinder liner protrusion (A) is the total sum of the liner flange thickness and shim thickness, minus the depth of the counterbore.

Cylinder Liner Protrusion (A)		
mm		in
0.13	MIN	0.005
0.18	MAX	0.007

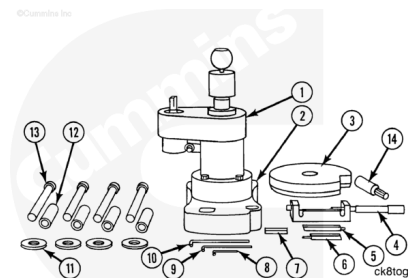


Machine for Oversize Liners.

Tools required for this procedure are listed below. They are available separately or in kits.

Liner Counterbore Tool Kit, Part Number 3377350

- 1. Liner Counterbore Drive Unit, Part Number 3377350
- 8. [0.9375 inch] Hex Key, Part Number ST-1168-57, or equivalent
- 9. [0.1563 inch] Hex Key, Part Number ST-1168-58, or equivalent
- 10. [0.1875 inch] Hex Key, Part Number ST-1168-60, or equivalent.



Liner Counterbore Salvage Tool Kit, Part Number 3375820

- 2. Base Plate, Part Number 3375827
- 3. Cutter Plate Assembly, Part Number 3375821
- 4. Micrometer Assembly, Part Number 3375826
- 5. Setting Standard, Part Number 3375828
- 6. Cutter Assembly (Part number depends on block type.)
Refer to following text
- 7. Depth Spacer Block (part number depends on block type.) Refer to following text
- 11. Plain washers, four, Part Number 3375830, or equivalent. [0.1875 inch inside diameter], [2.000 inch outside diameter] Standard Adapters, Part Number 3375829, or equivalent. [0.8125 inch inside diameter, 3.500 inch length].

The following is also required:

- Four cylinder head capscrews
- Drill Motor (10-ampere, 450 RPM, 19 mm [0.750 inch] Chuck).

The user of the following instructions **must** have inspected the cylinder block.

Note : Terms used in the procedure **must** be understood.

Note : After machining both the upper and lower counterbore inside diameter for oversized liners, you **must** machine the counterbore ledge.

These instructions cover the counterbore depth and the upper counterbore inside diameter machining at the same time for the following block.

Note : Cummins Inc. recommends that the accuracy of the tool be checked before machining the customer's block. If possible, machine a scrap block and check the accuracy of the cut.

If a scrap block is **not** available, adjust the tool to machine a size smaller than required. Check the accuracy of the cut. Perform final adjustments before machining to specification.

Note : The cylinder liner inside diameter is the same on oversize liners and standard liners.

Note : On all QSK19 blocks, the upper counterbore inside diameter **must** be no less than 0.025 mm [0.001 in] smaller, and no more than 0.076 mm [0.003 in] larger than the outside diameter of the cylinder liner flange diameter.

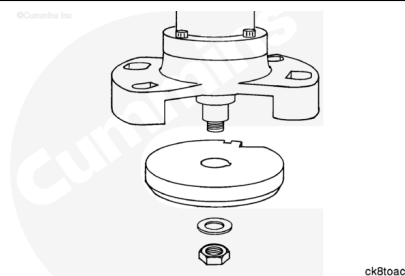
Install the base plate to the main housing. Install the four capscrews.

Torque Value: 41 n•m [30 ft-lb]

Position the cutter plate assembly with the tapered edge away from the housing. Align the keyway in the plate with the key in the shaft.

Install the cutter plate. Install the washer and nut.

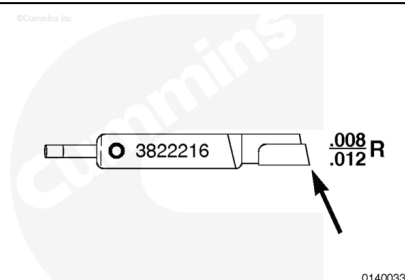
Torque Value: 41 n•m [30 ft-lb]



⚠ CAUTION ⚠

Damage to the cylinder block or interference with the cylinder liner will result if the wrong cutter assembly is used.

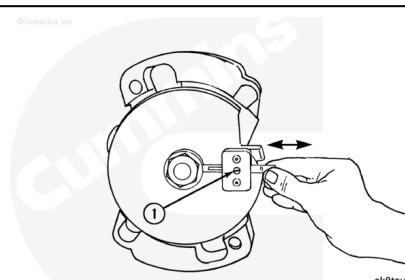
The illustration shows the cutter that **must** be used with the QSK19 cylinder block



Adjust the Set Screw

Put the correct cutter in the cutter plate assembly. Tighten the set screw until it is difficult to slide the cutter in the groove.

Remove the cutter assembly.



⚠ CAUTION ⚠

**The top of the cylinder block must be clean and free of burrs.
The tool must be level on the block surface to machine the counterbore inside diameter correctly.**

Measure the counterbore inside diameter and depth. Refer to Procedure 001-026 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-026-tr.html)

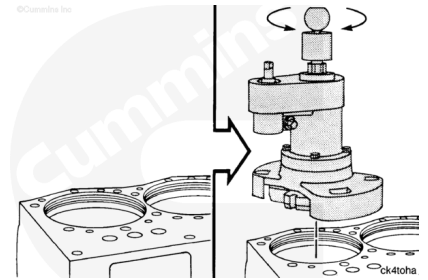
Install the Tool

To "LOCK" the feed mechanism, turn the knob on top **clockwise**.

To "UNLOCK" the feed mechanism, turn the knob **counterclockwise**.

"LOCK" the feed mechanism.

Put the tool on the block.



"UNLOCK" the feed mechanism.

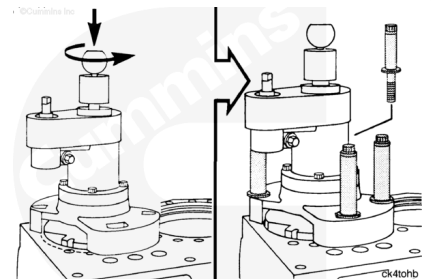
Lower the cutter plate until it touches the lower counterbore inside diameter. This will center the tool in the bore.

Be sure the base plate is flat against the block.

Rotate the base plate until the bolt holes align with the cylinder head mounting capscrew locations.

Install the four capscrews, plain washers, and standard adapters.

Torque Value: 47 n•m [35 ft-lb]



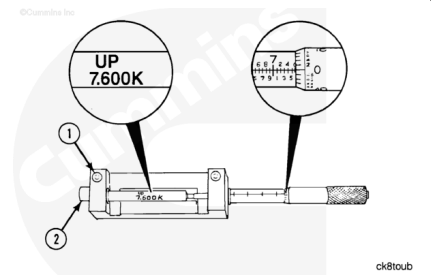
Adjust the Cutter Assembly

Use the setting standard, Part Number 3375828. Use the micrometer assembly, Part Number 3375826.

Put the setting standard in the micrometer, as illustrated. Measure the length of the standard. In this example, the micrometer indicates [7.600 inch].

If the reading does **not** agree with the value stamped on the setting standard:

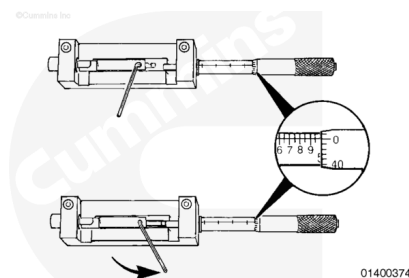
- Loosen the set screw (1)
- Move the end post (2)
- Tighten the set screw (1).



Set the micrometer to the correct dimension for the block being machined.

Put the cutter assembly in the micrometer, as illustrated.

Use a hex key to loosen the set screw on the cutter assembly. Move the spring loaded plunger until it contacts the micrometer.

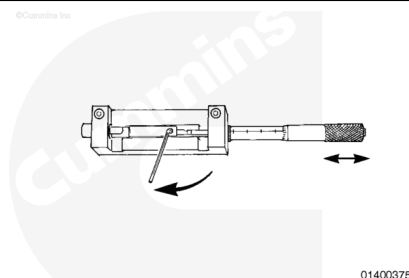


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Tighten the set screw.

Move the micrometer to check the setting.

Repeat the setting process until the cutter assembly is set accurately.



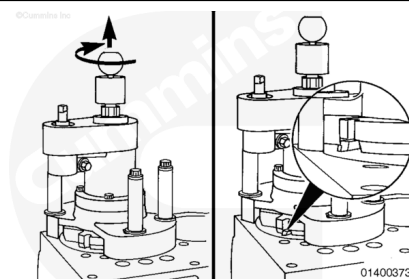
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Adjust Machining Depth

Raise and "LOCK" the drive assembly.

Put the cutter assembly in the cutter plate. It **must** extend a minimum of 6 mm [0.250 inch] over the edge of the counterbore.

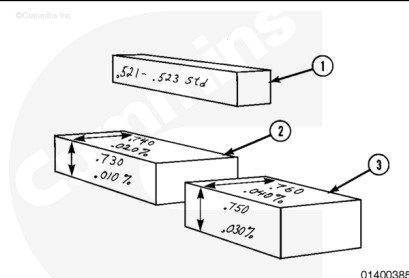
Tighten the top set screw for the cutter bit.



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On QSK19 cylinder blocks, use depth spacer block, Part Number 3376189.

Note : These depth spacer block is contained in the liner counterbore conversion kit, Part Number 3375820.



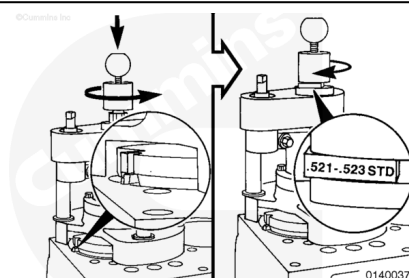
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"UNLOCK" the drive mechanism. Lower the cutter plate until the cutter touches the block.

Put the correct depth spacer block on the tool. The depth of the cut dimension **must** be visible, as illustrated.

Turn the stop collar until it touches the depth spacer block, as illustrated.

Note : The 13.3 mm [0.522 inch] depth spacer block is illustrated as an example only.



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Final Adjustments for Depth

The style of the block being machined determines the next adjustment.

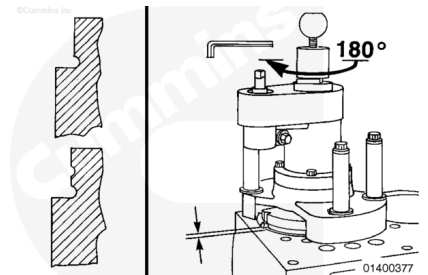
⚠ CAUTION ⚠

Do not cut the counterbore depth with this tool.

Follow these instructions for the QSK19.

Turn the depth stop collar 1/2 of a revolution (180 degrees) **counterclockwise**. The cutting assembly **must** be raised slightly from the counterbore ledge.

Use a hex key to tighten the set screw on the collar.



⚠ CAUTION ⚠

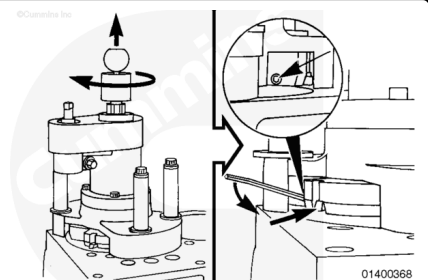
Be sure no dirt or chips are in the cutting assembly groove of the cutter plate.

Machine the Counterbore inside diameter

Raise the cutting plate assembly. "LOCK" the drive mechanism.

Push the cutting assembly into the cutter plate until it touches the shaft.

Use a hex key to tighten the set screw.



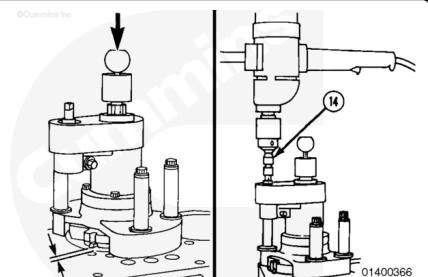
Note : Hold the drill firmly. The drill will be difficult to hold when the cutter bit initially touches the block.

Lower the cutting plate until the cutting assembly is 1.50 mm [0.0625 inch] above the block. "LOCK" the drive mechanism.

Install the universal drive (14) in the drill.

Note : The feed mechanism automatically controls the rate that the cutter is lowered.

"START" the drill. The cutter plate will free-wheel after the depth stop collar has contacted the drive unit.



"STOP" the drill after the cutter has free-wheeled for 5 to 10 revolutions.

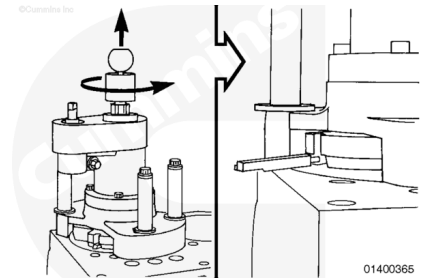
Remove the drill.

"UNLOCK" the drive mechanism.

Raise the cutter plate.

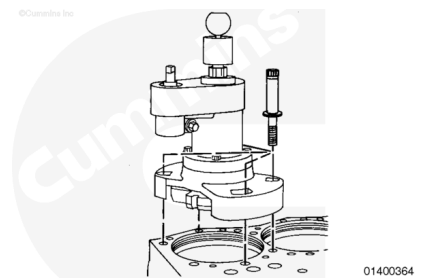
"LOCK" the drive mechanism.

Use a hex key to loosen the set screw. Remove the cutter assembly.



Remove the mounting capscrews, adapters, and washers.

Remove the tool.

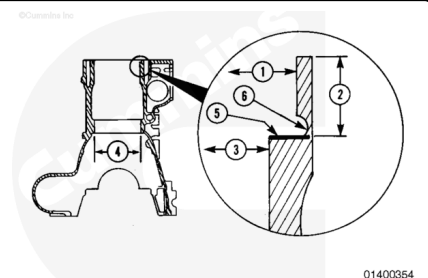


Use a hone stone. Remove burrs from the sharp corner of the inside diameter

Measure the upper counterbore inside diameter (1) in the location shown.

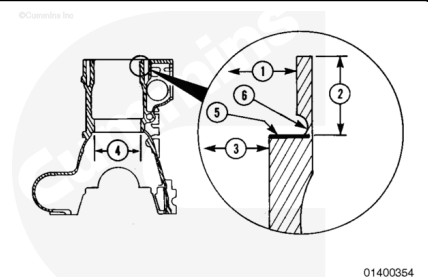
Machine the diameter again if the inside diameter is smaller than specification.

Machine the block for the next larger size liner when the inside diameter is larger than specification.



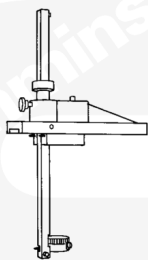
Check the concentricity between the upper counterbore inside diameter (1) and the packing ring bore (4).

Clean the cylinder block. Refer to Procedure 001-026 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-026-tr.html>)



Check the concentricity after the repair of the upper counterbore inside diameter or the packing ring inside diameter.

Use the concentricity gauge, Part Number ST-1252.



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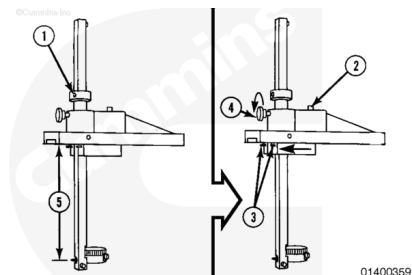
Adjust the contact pins (3) to be as close to the plate as possible, if necessary,

- Remove the set screw (2)
- Remove the plate
- Turn the plate over
- Install the plate
- Install the set screw.

Loosen the indicator arm set screw (1). Move the indicator arm until the distance between the contact pins and the indicator (5) is 254 mm [10 inch].

Loosen the contact plate set screw (2). Slide the contact plate as far toward the indicator arm as possible.

Loosen the adjusting knob (4).



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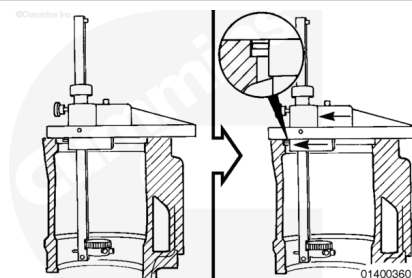
⚠ CAUTION ⚠

The upper counterbore and the top surface of the block must be clean.

Put the tool in the cylinder block liner bore to be checked.

Slide the contact plate until both contact pins touch the upper counterbore inside diameter.

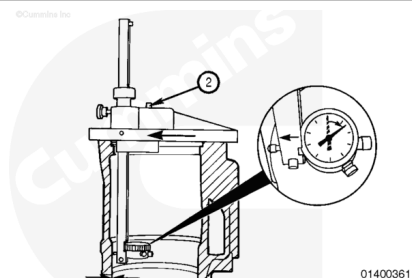
Note : The indicator must not touch the packing ring bore at this time.



01400360

Hold both contact pins in against the upper counterbore inside diameter.

Slide the base until the indicator tip touches the packing ring bore inside diameter. Continue to slide the base until the indicator needle has moved to 0.0064 mm to 0.254 mm [0.005 in to 0.010 in].



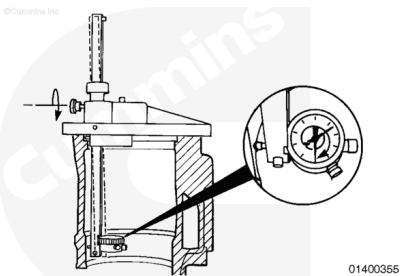
01400361

Hold the tool in place and tighten the contact plate set screw (2).

Hold the contact pins against the upper counterbore inside diameter. Turn the adjusting knob to move the indicator needle a minimum of one revolution. Turn the knob until the needle indicates "ZERO".

Slide the tool until the indicator is **not** touching the block. Slide the tool backward until both contact pins touch the upper counterbore inside diameter. The indicator **must** remain at "ZERO".

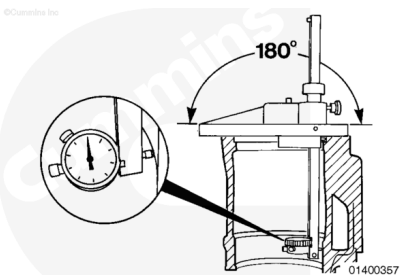
Repeat this step until the indicator continues to read "ZERO" after the tool is removed.



Turn the tool 180 degrees from the "ZERO" position. Slide the tool until both contact pins touch the upper counterbore inside diameter.

Read the indicator. The bore run-out is one-half of the indicator reading.

Example: Indicator reading of 0.2032 mm [0.008 in]. Actual run-out is 0.1016 mm [0.004 in].

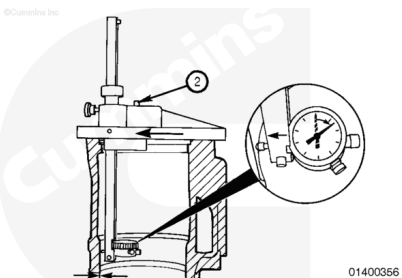


Liner Bore Run-Out

mm		in
0.130	MAX	0.005

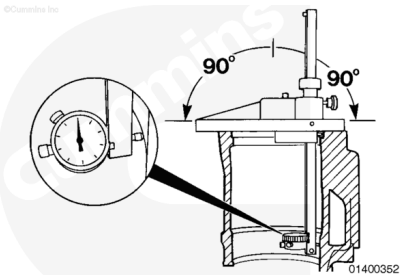
Move the tool 180 degrees backward to the original "ZERO" position.

Repeat the measurement procedure when the indicator does **not** read "ZERO".



⚠ CAUTION ⚠

Concentricity must be within specification. Engine damage will result when the liner is not seated correctly in the bore.



Turn the tool until the contact pins are 90 degrees
counterclockwise from the original "ZERO" position. Adjust the
indicator to "ZERO" position.

Read the indicator. Compare the reading to the specification.

Repeat this procedure until consistent readings are obtained.

**Note : The bore must be repaired when consistent
readings are not obtained. Reference the Alternative
Repair Manual, Bulletin Number, 3379035.**